

Technical Data Sheet

Date of issuing : 01.07.2025

Rev. No. 2

Product Name : Transformer Oil - TRANSOL HGX

Grade : IEC - 60296 (Edition 5.0 2020-06) - Inhibited, Type A (fully Inhibited high grade oils)

Sr. No.	Property	Test Method	Limits	Typical Values
1 - Function :				
1	Viscosity at 40 °C	ISO 3104	Max. 12 mm ² /s	10.8
2	Viscosity at - 30 °C	ISO 3104	Max. 1800 mm ² /s	1100
3	Pour Point	ISO 3016	Max. - 40 °C	- 42
4	Water Content	IEC 60814	Max. 30 mg / kg - for bulk supply	12
			Max. 40 mg /kg - delivery in drums/IBC	14
5	Breakdown voltage	IEC 60156	Min. 30 kV as delivered	40 - 60
			Min. 70 kV after laboratory treatment	>72
6	Density at 20 °C	ISO 3675	Max. 895 kg/m ³	864
	Density at 29.5 °C	ISO 3675	Max. 895 kg/m ³	859
7	DDF at 90 °C	IEC 60247	Max. 0.005	0.00035
2 - Refining / Stability				
1	Colour	ISO 2049	L 0.5 (less than 0.5)	L 0.5
2	Appearance	-	Clear, free from sediment and suspended matter	Clear, free from sediment & suspended matter
3	Acidity	IEC 62021-2 or 62021-1	Max. 0.01 mg KOH/g	0.0014
4	Interfacial Tension	IEC 62961 or ASTM D 971	Min. 43 mN/m	47
5	Total sulphur content	ISO 8754 / or ISO 20847	Max. 0.05 %	0.002
6	Corrosive Sulphur	DIN 51353	Not corrosive	Non corrosive
7	Potentially Corrosive Sulphur	IEC 62535	Not corrosive	Non corrosive
8	DBDS	IEC 62697-1	Not detectable (< 5 mg/kg)	Non detectable
9	Inhibitors of IEC 60666	IEC 60666	Inhibited Oil (I); 0.08 % to 0.4 %	0.38
10	Metal passivator additives of IEC 60666	IEC 60666	Not detectable (< 5 mg/kg) or as agreed upon with the purchaser	Not detectable
11	Other additives	-	-	NIL (Not added)
12	2 - Furfural & related compounds content	IEC 61198	Not detectable (< 0.05mg/kg) for each individual compound	Not detectable
13	Stray gassing under thermo - oxidative stress	Procedure in Clause A.4 (oil saturated with air) in the presence of copper	Non stray gassing <50 µl/l of Hydrogen (H ₂) <50 µl/l methane (CH ₄) < 50 µl/l ethane (C ₂ H ₆)	Non stray gassing Non stray gassing Non stray gassing
3 - Performance :				
1	Oxidation Stability	IEC 61125 Test duration : (I) Inhibited Oil : 500 h		
a)	Total acidity	4.8.4 of IEC 61125: 2018	Max. 0.3 mg KOH / gm	0.14
b)	Sludge	4.8.1 of IEC 61125: 2018	Max. 0.05 %	0.012
c)	DDF at 90 °C	4.8.5 of IEC 61125: 2018	Max. 0.05	0.0042
4 - Health , safety and environment :				
1	Flash point	ISO 2719	Min. 135 °C	148
2	PCA content	IP 346	< 3.0 %	0.52
3	PCB content	IEC 61619 / ASTM D 4059	Not detectable (< 2 mg/kg)	Not detectable


Authorised Signatory